

Blessing Chukwuma

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Education

Master of Public Health (MPH), University of Arkansas, Fayetteville January 2024 - December 2025

Relevant Coursework: Epidemiology, Statistics for Biologists, Research in Healthcare, Health Behavior, Health Theory

Final Project: "Exploring contraceptive counselling in Healthcare Settings using a Socioecological Model Approach"

Bachelor of Science in Zoology, University of Lagos Nigeria September 2017 – December 2021

Dissertation: "Efficacy of RTS S vaccine against Plasmodium falciparum strains in Lagos state, Western Nigeria"

Research & Professional Experience

Assessment Coordinator (Data Analytics & Reporting), University of Arkansas January 2026 – Present

- Improved the usability and reliability of institutional data by using SQL and Python to clean, transform, and analyze datasets supporting departmental research and campus-wide wellness program evaluations.
- Increased accuracy and consistency of institutional reporting by partnering with IT, clinicians, and administrative leadership to define and standardize key performance indicators, reducing ambiguity in data and campus space utilization metrics used for decision-making.
- Advanced evidence-based expansion of clinical services by leading an IRB-approved research project analyzing embedded clinician programs, producing findings presented to senior leadership and directly supporting a \$150K grant application.
- Accelerated reporting and operational insight by building 10+ Power BI dashboards tracking health, wellness, and service utilization across the university population, cutting report turnaround time from two weeks to four hours.

Data Analyst (Graduate Assistant), University of Arkansas, Fayetteville January 2024 – December 2025

- Strengthening evidence-based decision-making in student affairs by supporting the full assessment cycle—data collection, analysis, monitoring, and reporting—helping translate program outcomes into actionable insights that support student success.
- Advancing ethical and compliant research practices by assisting with IRB protocol development and submissions, ensuring institutional assessments and research activities meet regulatory and ethical standards.
- Expanding institutional analytic capacity by collecting, cleaning, and analyzing cross-unit data, responding to internal and external data requests with timely, decision-ready insights.
- Enhancing visibility of performance and trends by developing dashboards and executive summary reports, improving access to key metrics for leadership and program teams.
- Supporting a strong data governance and data literacy culture by contributing to standardized data definitions, best ethical practices, and data documentation, improving consistency, trust, and reuse of institutional data.

Summer Researcher (Population Health & Analytics), Community Clinic, NW Arkansas May 2025 - August 2025

- Addressed high rates of avoidable emergency room utilization by analyzing 244 patient records using Python, SQL, and SPSS, identifying key utilization drivers that informed initiatives projected to reduce unnecessary ER visits by 35%.
- Improved equity in access to care by applying geospatial analysis (QGIS) to map service gaps, guiding three targeted outreach initiatives reaching 500+ minority and underserved patients.
- Enabled data-driven care prioritization and resource allocation by building risk stratification models, empowering clinical teams to focus on high-utilization patients and informing facility placement decisions in underserved zones.

Data Analyst Trainee, Helium Health, Nigeria March 2023 – December 2023

- Addressed gaps in patient engagement and service delivery by cleaning, processing, and analyzing EMR data from 30+ health facilities, identifying operational bottlenecks that contributed to a 30% improvement in appointment adherence within six months.

- Enabled proactive quality management by hospital administrators through implementing time-series analyses across five core health and operational metrics, supporting early detection of service quality declines and preventing patient backlogs in two regional facilities.
- Improved reliability of donor and clinical reporting by collaborating with data, clinical, and grants teams to refine health metric definitions and validation rules, ensuring repeatable, audit-ready data quality for internal decision-making and external reporting.
- Supported expansion of digital health services in underserved communities by contributing data analysis and monitoring support to a Merck for Mothers–funded initiative, extending EMR adoption across rural health facilities and strengthening access to maternal health services.

Research Assistant, Nigerian Institute of Medical Research (NIMR) Foundation November 2021 – March 2023

- Strengthened grant competitiveness and funding readiness by producing statistical analyses and evidence for grant proposals, directly contributing to securing \$50K+ in research funding for maternal health and infectious disease programs.
- Built research capacity among early-career scientists by delivering grant writing and research development training to 100+ researchers, increasing successful funding applications by an estimated 70% and enabling the launch of five high-impact research projects in Nigeria.
- Supported strategic oversight of multi-year research programs by compiling progress reports and performance summaries, informing leadership decisions across five active research projects.
- Enabled smooth research and organizational operations by coordinating administrative functions including budgeting, tax documentation, and office setup, ensuring compliance and operational continuity for grant-funded projects.
- Expanded the Foundation's visibility and engagement by creating research-focused content and managing social media platforms, amplifying dissemination of research outputs and stakeholder awareness.

Public Health Research Assistant, Nigerian Institute of Medical Research November 2019 – January 2020

- Supported national efforts to reduce malaria and neglected tropical diseases by conducting rigorous literature reviews and statistical analyses for multi-site studies, producing evidence that informed Ministry of Health policy briefs and public health decision-making.
- Strengthened the quality and reliability of national malaria surveillance data by designing, piloting, and deploying digital survey tools (KoboToolbox, Google Forms), ensuring 100% compliance with IRB and ethical standards and improving data integrity used to monitor disease spread.

Volunteering

Health Strategy Intern, National Association of Chronic Disease Directors, Atlanta Georgia January 2025 - Present

- Currently leading needs assessment projects on chronic disease prevalence and care barriers, shaping grant proposals to target underserved countries in 3 sub-Saharan Africa states.
- Developing a HPV vaccination expansion plan projected to raise coverage rates by 20% in Malawi and Eswantini over five years.

Public Health Volunteer, Health & Medical Community Development Group, Nigeria March 2022 – February 2023

- Led community outreach programs reaching 60+ market traders with preventive healthcare education, increasing awareness of disease prevention practices by 45%.
- Evaluated program outcomes using collected data, refining strategies that improved healthcare access for underserved traders.

Project Manager, Youth Against HIV Discrimination, Nigeria August 2022 - July 2023

- Managed a school outreach program aimed at improving HIV knowledge and reducing stigma among early adolescents across 10 schools.
- Facilitated participatory training sessions to build students' understanding of prevention behaviors and promote positive health attitudes.
- Tracked implementation progress, identified barriers to engagement, and communicated weekly updates to project leadership.

Research Publications Under Review

- Women and diabetes: the impact of menstruation on glycemic control: A Scoping Review. Kylie Lovett, MPH, CHES; Blessing Chukwuma, MPH.
- Contraceptive coercion, method satisfaction, confidence in use, and switching intentions among women of reproductive age in the U.S. Shristi Bhochhibhoya, PhD.; Blessing Chukwuma, MPH; Kylie Lovett, MPH
- Proximity to School-Based Health Centers and Preventable Emergency Department Visits: An Exploratory Study: Abbie Page PhD; Blessing Chukwuma, MPH, Lisa Stafford, MHA MSc
- Therapeutic Strategies for Alzheimer's Disease: A Narrative Review Tracing the Evolution from Symptomatic Treatment to Multimodal Precision Medicine. Samuel Kakraba, PhD, Joseph Asante Jnr., PhD, Enoch K. Larrey, Desmond Yemeh, PharmD, Mamadou Sangare, PhD, Edmund F. Agyemang, Benjamin Tommy Bavug, Godsway Believer Gbeze, Blessing Chukwuma, MPH.

Professional Development

Data Science Intern, Zion Tech Hub, Remote

September 2025 - December 2025

- Cleaned and engineered features for a diagnosis machine learning prediction task.
- Built a machine learning project to predict hypertension using health survey data.
- Applied convoluted neural networks (CNN) to train and x-ray image diagnosis data.
- Built and deployed machine learning models using a Streamlit app.

Guest Lecture

PBHL 53103 - Public Health Evaluation 2025: Statistical Analysis of Research Data Using SPSS.

Conference Presentation

- "Women and diabetes: The impact of menstruation on glycemic control - A Scoping Review", **American Public Health Association**, Washington DC, USA November 2025
- "Women and diabetes: The impact of menstruation on glycemic control - A Scoping Review", **Health and Human Profession Research Day**, Fayetteville, Arkansas, USA April 2025
- "Proximity to School-Based Health Centers and Preventable Emergency Department Visits: An Exploratory Study ", **American Academy of Health Behavior**, Austin, Texas, USA. March 2026
- "Contraceptive coercion, method satisfaction, confidence in use, and switching intentions among women of reproductive age in the U.S.", **American Academy of Health Behavior**, Austin, Texas, USA. March 2026
- "Association of Social Determinants of Health on provider-based contraceptive coercion across races", **American Public Health Association**, San Antonio, Texas, USA November 2026

Leadership

President, National Association of Zoology Students, University of Lagos, Nigeria October 2019 – November 2021

- Led a team of 7 in organizing career development workshops for over 300 students, increasing department-wide engagement by 30%.
- Spearheaded initiatives that connected students with industry professionals, fostering academic and career

Skills

- Data Analysis & Visualization: Proficient in Power BI, Excel, SPSS, SQL, Python; familiar with SAS and R.
- Business Development & Strategy: Grant writing, market research, competitor analysis, and proposal development.
- Research & Consulting: Expertise in hypothesis-driven problem-solving, market research, and strategic analysis.
- Communication: Proven ability to present complex data and insights to both technical and non-technical audiences.
- Technical Proficiency: Advanced in Microsoft Office Suite (Excel, PowerPoint, Outlook)

Certifications

- SAS Certified Specialist: Base Programming
- DataCamp Associate Data Analyst
- Lean Six Sigma Green Belt

- Scrum (Agile) Project Management

Award

2025 GPSC Outreach Award: Recognized for leadership in community-engaged research and contributions to advancing health equity through data-driven outreach programs.

Others

Judge – Young Data Professionals Nigeria Hackathon (Healthcare and Education sections)

Mentor – Knowledge Tech Hub (free data analysis and science training for new folks)

Mentor – Guidrs (company that helps people get into graduate school – healthcare specified)